

A Mixed-Method Case Study: Medical Practitioner Sensemaking in the Context of Practice Analytics

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ABSTRACT

There is a large amount of data generated by the healthcare system that can be leveraged for secondary purposes. Here we present a case study, centred around medical practitioners, that explored how they interact with such data to inform their professional development and learning. We used a mixed-methods approach (interviews and surveys) to glean rich insights into their sensemaking and present considerations that can subsequently support this. These human-centred considerations include the use of “frames of reference”, presentation of “cases of interest”, significance of data attribution, factors that impact sensemaking (i.e. affect and prior experiences), and we advocate for iterative co-design. Alignment to such considerations would ensure the analytic tools are developed in a way that complement practitioners’ needs, and are meaningful to both their clinical practice and professional development.

CCS CONCEPTS

• **Human-centered computing** → *HCI theory, concepts and models*; • **Applied computing** → *E-learning*; Collaborative learning; Health informatics; Psychology.

KEYWORDS

Case Study, Continuing Professional Development (CPD), Interview, Mixed-Method, Practice Analytics, Sensemaking, Survey

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1 INTRODUCTION

There is an increasing amount of data being generated within healthcare [8]; an acceleration driven largely by the digitisation of healthcare systems and processes. This includes the use of electronic medical records, electronic health records, electronic patient administration systems, and electronic claims and billing services. With this comes a wealth of data sources that have the potential to be re-purposed to generate new insights on different aspects of healthcare. Whilst some more obvious examples include using the data for clinical registries, trials, and medical research, an example that has received less exploration is using this data to support professional learning and development. This concept is central to research within the field of practice analytics [18, 19, 27].

The research area of practice analytics complements international shifts towards more data informed professional learning and development [18, 19]. Increasingly, medical regulators and professional bodies are promoting electronic/digital learning tools within their professional frameworks [12, 14, 24, 28]. A notable example is in Australia where medical practitioners are expected to spend a minimum of 25 hours per year reviewing data associated with their clinical performance/outcomes [24]. This forms part of their yearly Continuing Professional Development (CPD) requirements, and is therefore mandatory in order for them to retain their license to practice. As such, it is crucial that reviewing such data is meaningful, actionable, and enhances clinical practice; so it does not become an additional burden, or a “tick-box” exercise. As such, there is a call to explore medical practitioner sensemaking in this context to ensure that the practitioners have tools and reporting centered around their needs [33].

Here, we explore medical practitioner sensemaking using a case study approach at one innovating hospital within Victoria, Australia. Such approach is of particular significance given the infancy of both the research area of practice analytics and also the lack of routine and standardised data sharing with practitioners within hospitals. We conclude with clear contributions in the form of human-centered considerations for implementation of analytical tools to ensure that medical practitioner sensemaking is considered. The hospital included within this case study remains one of the only hospitals within Australia to routinely share data with its practitioners, and thus its insights are both novel and have implications for implementation across future hospitals and services.

2 BACKGROUND

2.1 Practice analytics

Practice analytics aims to re-purpose the plethora of data within healthcare and leverage it to support practitioners' performance reflection and CPD. Whilst other audit processes exist that review clinical performance [3, 17], in general they take a top-down approach by beginning with certain indicators that require measurement. Such approaches are often very specific in their reporting. Where practice analytics differs is it seeks to take a bottom-up approach. Instead of beginning with the indicator and then generating data on that specific indicator, it begins with providing a practitioner with a range of data associated with *their* clinical performance allowing them to explore and generate insights useful for *their* development, learning, and clinical practice enhancement. It is both more self-regulated and exploratory in contrast to the aforementioned audits and requires practitioners to individually explore the data to generate insights and make inferences about their performance. Moreover, it requires them to "translate" such data into learning and subsequently, their clinical practice. Given that such process places significant emphasis on the practitioner regulating their *own* learning and review of the data, it important that research explores exactly how practitioners can be supported to ensure that the data is both meaningful and actionable for their learning and clinical practice. This would ensure that tools are centered around practitioners needs and increase the likelihood of success. Given that already established audit processes report varied impact and responses from practitioners [17], an exploration into medical practitioner data sensemaking, for this more exploratory context, is essential [33].

2.2 Continuing Professional Development (CPD)

The primary objective of CPD is "*to support quality, lifelong learning for doctors that is relevant, effective and evidence-based*" [24]. Within Australia, many different activities constitute towards CPD requirements allowing the medical practitioner flexibility to tailor their development around their individual needs and scope of practice. This flexibility is reflected within the new professional performance framework as, whilst medical practitioners must complete 50 hours of CPD activities a year, 25 of those are to be spent broadly "reviewing performance and measuring outcomes" [24]. Emphasis is placed on performance reflection through activities such as "*audit focused on participant's own practice*" and "*self-evaluation and reflection*". The volume of data within healthcare can support such reflection, generating prompts and insights that otherwise may not be realised.

Reflecting on clinical performance or "reflective practice" is common within healthcare and is a concept that has long been ingrained within the health professions. Seminal work by Schon [30] regarding the "reflective practitioner" differentiates between reflection-*in-action* and reflection-*on-action*. The first involves the process of reflection during an activity, e.g., a surgeon iteratively reviewing their performance whilst in the operating room. The latter involves reflecting on an activity after it is complete, e.g., thinking

back and reviewing a past experience. Integrating data, particularly re-purposed data that was not initially intended to prompt reflection-*on-action*, is likely to be a new concept for most. Exemplified by its continued inclusion within CPD [24], whilst traditional reflective practice positively impacts clinical outcomes and enhance clinical practice, at present no primary research has explored how medical practitioners "make sense" of this re-purposed data performance for reflection and CPD purposes. As such, we address the call for research that explores medical practitioners sensemaking in this context [33].

2.3 Sensemaking

Sensemaking involves extracting cues from a situation, and through cyclic processes of interpretation and enactment, assigning meaning to it [22, 29]. It is a highly subjective process, because (i) the cues individuals attend to can differ between individuals [29], (ii) meaning is shaped by individual experiences and prior knowledge [32], and (iii) the process is impacted by many factors, inclusive of context and emotional state [29]. As a result, different individuals may take away a different meaning despite viewing the same stimulus. Research into the phenomenon spans many different fields inclusive of organizational psychology [32], HCI [26], information sciences [10], and learning analytics [6]. As a result, many rich, but varied, perspectives/theories exist about such phenomenon [22]. Recent publications have taken steps towards aggregating a variety of literature to generate exhaustive and synthesised definitions [22], frameworks [29], and methodologies [9]. The latter of which is of key significance given the challenges in observing sensemaking which contributes to the variety of different definitions and theories. It remains challenging to measure sensemaking given it is an internal process that is therefore difficult to observe. However, attempts have been made either through the development of novel metrics that make inferences about sensemaking such as immersive environments [34] and annotations [16], and also through more traditional methodological approaches such as interviews [1], think-aloud techniques [21], observations [23], and surveys [2, 9].

First to consider is the development of novel methods to explore sensemaking. Embodiment has been used to understand and subsequently guide sensemaking. This process involves representing concepts as virtual objects and subsequently observing how users both interact with, and openly explore the different scenarios. This has been done through visualisations [7] and also virtual reality [15]. Zhang et al. [34] expand on this further and propose "*a concept of embodied provenance*" using an immersive environment. Their use of the three-dimensional environment generated design considerations that support individual sensemaking efforts through processes of recall, reproduction, annotation, and sharing. Whilst a prototype, the results presented by Zhang et al. [34] have implications for designing tools that support, facilitate, and promote sensemaking. In addition, development of specific tools has allowed for explorations into more specific and applied sensemaking such as "*Catseye*" that used annotations to make sense of coding [16] and "*SensePath*" that used analytical provenance to automatically make inferences about sensemaking within browsers [25]. Such approaches generate rich, but highly contextual, accounts of sensemaking.

More traditional methodological approaches to observe sensemaking have also been used and include the use of observation [23], think-aloud [21], and surveys [2, 9]. Of note, De Liddo et al. [9] expanded on the work of Alsufiani et al. [2] to develop a survey based on a 9-factor framework of sensemaking. Whilst published within the HCI community, the survey items were formulated based on sensemaking literature that was inclusive of multiple research disciplines. The use of a survey brings key benefits to understanding sensemaking in specific contexts but potentially on a larger scale. Others have also triangulated across methodological approaches to generate rich accounts of sensemaking [21]. Given the challenges in observing sensemaking, this case study triangulates the findings of both a survey *and* interviews, to explore medical practitioner sensemaking in the context of practice analytics. The study both explores the sensemaking process that is enacted in this context, and presents considerations, grounded in the findings, to support sensemaking that is effective and conducive to learning and development.

3 METHODOLOGY

3.1 Case study setting

A case study approach was taken to understand medical practitioner data sensemaking at one private hospital in Victoria, Australia. This site was of interest as it routinely shares data with practitioners on their clinical performance. It remains one of the only hospitals to share such data routinely in Australia. Given the innovation of this hospital in relation to data sharing, a case study approach was taken in order gain an in-depth understanding into the process of medical practitioners' data sensemaking. Triangulation across different methods (survey and interview) allowed for a multifaceted approach to data analysis and insights that generated a well rounded exploration into sensemaking. Ethical and governance approval for the survey and interview approaches was obtained from Monash University Human Research Ethics Committee and the hospital site (IDs: 31962 & 24779 and 06-07-04-22 & 08-03-09-20, respectively).

3.1.1 Quarterly Data Reporting. Medical practitioners within the hospital received a quarterly report that outlined their clinical performance (Appendix A.1.1). The data was presented in a tabular format within an Excel spreadsheet and shared via email. The data represented the previous quarter, in three formats. The first provided an aggregated summary of the speciality group performance, the second presented all of the planned and emergency procedures (grouped by the type of procedure), and the final format presented "cases of interest" that were manually identified as being worthwhile for review. Throughout the reports, the same indicators were used as measured of clinical performance.

3.2 Recruitment and respondents

3.2.1 Survey recruitment and respondents. The survey population was medical practitioners¹ who received a copy of the quarterly data report referenced above (N=226). A link to the survey was

shared via email in conjunction with the quarterly report. All practitioners who received the quarterly report were invited to take part in the survey. A total of 44 respondents began the survey of whom five did not fully complete the survey; thus, their data were removed. This provided a 17% response rate (39 complete responses, Appendix A.1.4). Given this response rate and that a variety of speciality groups and demographics were represented, the sample was considered representative of the population.

3.2.2 Interview recruitment and respondents. The interview population was individuals who had experience with handling the quarterly report data. Whilst this predominately included practising practitioners, there were also additional people within the hospital who handled such data (e.g. medical directors or health informaticians). Respondents were recruited using convenience sampling methods. Second author, DR, reached out to personal contacts at the site inviting them to be interviewed. The final sample included 11 respondents (10 medical practitioners and 1 non-practising medical director²). In order to protect the anonymity of the quoted respondents from this specific and small population, a full demographic breakdown is not provided but the sample included representation from a variety of speciality groups.

3.3 Data collection

3.3.1 Survey data collection. The first data collection method involved a survey (Appendix A.1.2). The survey link was shared electronically in conjunction with the quarterly data report and was hosted on Qualtrics. The first section of the survey sought demographic information from the respondent e.g. speciality group, length of practice, and a self-reported measure of how confident they were in extracting meaning from data. The second section was dedicated to exploring the process of data sensemaking; how the individual practitioners were "making sense" of the data that was presented to them. These questions were adapted from De Liddo et al. [9], who proposed a 9-factor framework for sensemaking based on an extensive review of the literature. Thus, the questions explored nine different elements of the sensemaking process (listed within Table 1 and labelled as SM1-SM9). The items were only adapted to account for the context (Appendix A.1.2). The survey items were grouped thematically for this context as outlined in Table 1. The final section collected qualitative and quantitative feedback on the quarterly data report. Five-point Likert scales were used to assess data usefulness, validity, peer comparisons, comfort in sharing, concerns, and links to CPD.

3.3.2 Interview data collection. The second data collection method involved interviews. Two interview protocols (Appendix A.1.3) were developed, one for those currently clinically practising, and one for those who were not clinically practising. Questions within the protocols centered around existing data processes and elicited commentary on data presentation, types of data included, how the data was being used, and any barriers/concerns surrounding the use of the data. Interviews lasted between 17 and 50 minutes and were conducted virtually by EWW. The interviews were all transcribed

¹To clarify terminology, in Australia, a "health practitioner" is a broad term and encompasses a range of different types of practitioners e.g., dentists, nurses, pharmacists, and physiotherapists. A "medical practitioner" is considered a specific *type* of health practitioner e.g., a physician or surgeon.

²This medical director was a fully trained medical practitioner, however was not clinically practicing at the time of interview.

Case study theme	9-factors of sensemaking proposed by De Liddo et al. [9]
(1) Performance reflection	Reflection (SM1)
(2) Generation of insights and information	Insights (SM2) and Focus (SM3)
(3) Performance reconstruction	Argumentation (SM4)
(4) Data-driven evidence	Explanation (SM5), Assess facts & evidence (SM6), and Distinguish (SM7)
(5) Assumptions	Assess assumptions (SM8) and Change assumptions (SM9)

Table 1: A table to outline the 9-factors of sensemaking and how these map to the themes presented within the results.

manually and verbatim, and respondents were anonymised during this process.

3.4 Data Analysis

The results from the survey and interviews were thematically synthesised [4] to generate a rich case study account of sensemaking in this context. To achieve this, descriptive statistics are presented to represent the survey results. Quotes from the survey comments and the interviews substantiate the findings³. To further support the triangulation of these methodologies, the results and discussion sections are integrated in order to situate the findings within the specific case study context, the existing CPD processes, and the existing literature.

4 RESULTS AND DISCUSSION

The results are thematically grouped. Additional response distributions and quotes to support findings are in Appendix A.2.1 and A.2.2, respectively. In order to contextualise the “sensemaking” exploration, first a detailed account of *data-driven practice reflection as a constituent of CPD* is provided before the sensemaking themes (1-5).

Data-driven practice reflection as a constituent of CPD

The data was deemed useful for learning and development. Within the survey, 72% of respondents felt that reviewing it should constitute towards CPD. This was further emphasised within the survey comments: *“worthwhile and a greater potential for change [compared] to other CPD activities”* (survey). Learning and improvement, more generally, was also referenced: *“it just keeps us in check ... it’s very easy just to ... do your own thing and not sort of be part of a group, or the real world. It’s important to ... adjust things and discuss things and you learn and improve”* (interview). Such findings support the premise of practice analytics i.e., re-purposing of data within healthcare to guide performance reflection [5, 19], and add weight to its inclusion with international professional development frameworks [12, 14, 24, 28].

Whilst most saw the benefit in using this data to support performance reflection and CPD, others referenced challenges surrounding meaningfulness and interpretation: *“I couldn’t understand the reports ... I didn’t understand the terms being used and I couldn’t understand what was my data and how it related to the other data, or what the group [data] related to. [Altogether] confusing”* (survey). This highlights that, whilst there may be an *opportunity* to re-purpose data for performance reflection and CPD, integration

to practice is not yet fully aligned with such vision. Research and policy surrounding effective implementation is in its infancy: *“the information about patient outcomes after surgery ... is often not always fed back to the clinicians, it probably should be, but it’s not in many cases”* (interview).

Likely due to such infancy, practitioners reported varied levels of confidence in extracting meaning from the data. Whilst the majority felt that they were either “completely confident” (18%) or “fairly confident” (44%) in extracting meaning, the remainder reported that they were only “somewhat confident” (25%), “slightly confident” (5%) or “not confident at all” (8%). Respondents acknowledged individual differences contributed to varied interpretation, including clinical training: *“they’re now taught to interpret data ... interpreting data and patient outcomes and patient experiences just germane to what we do”* (interview) and, somewhat conversely, large data volumes: *“that’s the challenge because everyone’s got a lot of data these days and people just don’t know where to look”* (interview). Others expressed little benefit in reviewing the data and instead were critical of the data quality (*“not risk stratified, no consideration given to patient complexity, inaccurate information related to outcomes and emergency/elective status”* (survey)), its validity as a measure (*“if you’re doing slick, good, cheap surgery, that’s great, but you need to know what that means down the track”* (interview)), and reservations about data source (*“we know that administrative data ... it’s weaker in many areas ... it just doesn’t have the input from the clinicians”* (interview)). For these respondents, their sensemaking efforts centered around *evaluating the data* which potentially distracted them from taking away insights that were meaningful or productive in terms of *evaluating their own clinical practice*. This highlights an opportunity to explore and develop tools that promote sensemaking that is conducive for learning, development, and enhances clinical practice. The process of sensemaking is inevitable when practitioners are presented with data, but the their sensemaking must be guided down a constructive path of learning and development. The remainder of the results focus on medical practitioner sensemaking, thematically grouped by the survey factors. An understanding of which will support the development of reporting in practice analytic tools that encourages sensemaking that is conducive for *“support[ing] quality, lifelong learning for doctors that is relevant, effective and evidence-based”* [24]; a primary objective of CPD.

(1) Performance reflection

The first sensemaking factor was reflection, proposing that during the sensemaking process, individuals reflect on prior experiences to make sense of the present [9]. Many of the survey respondents felt that the data helped them reflect on their performance in a deeper

³Additional quotes to support the results can be found in Appendix A.2.2.

way (53.85% “agree” and 2.56% “strongly agree”, SM1). Respondents also referenced using this data to support reflection individually: *“I think for the majority of [specialists] it probably is an individual type thing ... let’s say if their results are slightly sub optimal ... they probably just want the opportunity to actually self-correct rather than necessarily advertise it to their peer is sort of things”*, and as a group: *“they’re useful ... We’re together and we’ll pull a bit of information ... that we will discuss as a group”*. A similar sentiment was reflected within the survey; 67% indicated comfort in sharing this data with their colleagues, the remainder had some reservations.

Some respondents explicitly referenced data reflection and the impact it had on their development or practice: *“this is what CPD is about – reflection on your data is integral to improving your performance”* (survey). Whilst the results presented here infer that reflection may be a part of the sensemaking process, this may be due to the nature of the task i.e. reviewing historic data primes practitioners to think historically. Despite this however, given the data is intended to prompt reflection, it is encouraging that it seemed to do so.

(2) Generation of insights and information

Two sensemaking factors (insights and focus) [9] were associated with this theme. Here, two survey items (SM2 and SM3) explored whether the data provided practitioners with new insights and helped them focus. Whilst the majority found reviewing the data useful (82%), less than half felt that it was a valid measure of their performance (46%). Reservations centered around small surgical volumes (unrepresentative), accuracy, and validity. These reservations may have contributed to the varied response distributions for SM2 and SM3. Whilst 23% of respondents agreed that the data generated *“unexpected insights”* and 28% felt as though it allowed them to *“focus on different aspects of [their] performance”*, 44% and 36% respectively remained impartial to such sentiments.

In many cases, the insights generated by the data aligned with practitioners own perception of their clinical performance: *“it is a nice confirmation that all is actually going well with regards to my operating”* (survey). Not only does this serve as both positive reinforcement and promote a culture of sharing feedback (not just negative), it also highlights that practitioners currently lack visibility of these data-driven insights. This is less of a concern when the quality of care is high, but does highlight the opportunity that practice analytics seeks to address in providing practitioners with objective measures of their clinical performance. Whilst shifts towards data-informed performance reflection and CPD is a regulatory step towards implementation, hospitals may currently lack both the infrastructure and knowledge surrounding effective implementation. Moreover, this case study was conducted at an innovating hospital that already shares data with its practitioners. The majority of hospitals are not at this stage: *“when we first put these out ... de-identified physician level outcome data, it was probably the first time they had seen it at [hospital] ... the first time we’ve seen that sort of level of data in any health service”* (interview). Practitioners have even been labelled as *“flying blind”* given their lack of access to meaningful data to review their clinical performance [31].

Many practitioners (60% within the survey) reported interest in the insight gleaned from peer comparisons: *“it prompted me to have*

a look at what I was doing and how I compared to my peers” (survey). This may have given practitioners a “frame of reference” which is well reported to guide sensemaking in other contexts [21]. Even if practitioners become aware of their own performance, the comparison provides context. Furthermore, deviations from colleagues generates interest: *“we don’t like being outside the norm, the flock. We all like to be with the flock, so if anybody’s outside the flock, that’s, it’s powerful information”* (interview). Innately, practitioners do not want to be considered an outlier, and whilst this can drive change, this culture needs to be managed sensitively as some respondents raised cultural concerns: *“some of my peers are hypercritical in a non-constructive way”* (survey). Whilst labelling outliers could be confronting initially, if managed carefully, they may provide the most insight for practitioner learning and development: *“I would say you do want outliers because, and that outlier probably needs to be looked at you know. Why are they an outlier? Is it explainable, was it preventable non-preventable ... the outliers could be quite informative about a rare event that’s happened, and it might inform a change in practice ... outliers need to be looked at”* (interview). Whilst there is benefit in reviewing all aspects of clinical performance using data (i.e. increased awareness), if the quality of care is high, the practitioner may not need to change their practice. The presentation of data outliers (or “cases of interest”), however, may highlight areas of practice that need to change. These insights may have been invisible if the data was not shared; it also supports the generation of insights within the data sensemaking process.

(3) Performance reconstruction

The fourth sensemaking item (SM4) asked if the data helped practitioners reenact performance. The response to this survey item was mixed with many (36%) providing a neutral response. This may be due to negative wording in the question causing confusion and polarised results: 20% felt it helped reconstruct performance, 44% opposed this. The qualitative data indicates that this may have been because the data was too broad for their individual needs: *“it’s useful, but it’s very broad brush, would be nice ... to get a more individualized report ... more actionable”* (interview). The responses to this question may reflect the infancy of practice analytics and using data to prompt reflection. This may be the first time practitioners have used data to reflect on their performance and therefore they may struggle to connect it with their clinical practice. This is emphasised when practitioners requested more granular patient detail: *“the data are limited ... no patient names ... which makes interpretation somewhat difficult”* (survey). Whilst practitioners may require additional scaffolds or granularity to aid the reenactment, this element of sensemaking is crucial to support performance reflection and CPD. The practitioner uses their clinical knowledge and experience to translate data from being an abstract data point, to being a patient, a human life, that the practitioner impacted.

Furthermore, respondents referenced needing more information and time to contextualise the data. This infers sensemaking involves triangulation across different data (and/or discussions) to assign meaning useful for reflection. The following quote demonstrates such contextualisation and shows how practitioners interrogate data in a question-driven way: *“show them the Excel spreadsheet and it’s not going to mean a lot, so actually talking about it in detail*

... accessing the medical record to then make it a useful discussion ... why did this particular complication happen in this particular patient?" (interview). Given practitioners preferences for granular detail, implementation should consider individualised reporting, drill-down functionality, and/or case-based learning. These tailored approaches give flexibility to practitioners. Some value an individual report and the additional scaffolds that this offers, whilst others value the ability to flexibly drill-down and openly interrogate and explore the data to seek their own insights. This is referenced in the following quote: *"pick ... a dozen cases ... where you'd really want to drill-down on what happened ... what can we do to make sure that we do better next time"* (interview) and *"we can't ... press a button ... and we send it to them ... we need to analyse it and then pose more curated questions that engage them really, rather than passive ... assimilation of data"* (interview).

(4) Data-driven evidence

Three factors within the original survey [9] were thematically linked to data-driven evidence. First, whether data explained performance (SM5), second, whether the data evaluated facts and evidence (SM6) and third, its ability to help distinguish performance (SM7). Most survey respondents felt the data provided them with explanation of their performance (SM5, 38% "agree") and with ways to evaluate their performance (SM6, 51% "agree" and 3% "strongly agree"). Fewer survey respondents felt it helped them distinguish their performance (SM7, 8% "strongly disagree" and 28% "disagree" – negatively worded question). Whether the data generated objective evidence may be linked to perceived data validity i.e. if it could be accurately mapped to clinical performance and if it was a useful metric to reflect on. In order for the data to be meaningful for learning and development, practitioners must attribute it to their performance [33]. There is evidence that practitioners did not link the data to their performance: *"clinically completely irrelevant ... apples and oranges in many cases ... usual witch hunts ... because it is fundamentally meaningless"* (survey). To mitigate the risk of practitioners feeling such sentiment, they should be included within the development of the data reporting and practice analytic tools: *"sometimes the KPIs are decided by people who don't work on the front lines ... they must make sense to the clinicians otherwise they're pointless ... you're not going to ... change anything unless the people who are going to change it are involved in defining what it ... what matters"* (interview).

Including practitioners within the design of practice analytics has many benefits. It ensures that the data-driven evidence is meaningful for reflection and relevant to clinical practice, and also practitioners can understand the underlying motivations. At present, quotes jump towards labelling the data as an audit and/or assume negative affect: *"pointless stats ... makes no point"* (survey). This data is not intended to be punitive or as a label for poor performance, it is instead intended to support practitioners. As outlined within the related works, practice analytics places emphasis on practitioners independence and autonomy and seeks to provide them with the tools to prompt meaningful reflection. It complements the flexibility already with CPD frameworks [24], by equipping practitioners with data they need to meaningfully reflect on. For some, changing mindsets from targeted *"witch hunts"* (survey) to: proving useful

information for self-exploration and development, will require a shift in mindset towards data, and that will take time. Moreover, such negative affect has already been shown fundamental impact sensemaking in this context [33].

(5) Assumptions

The final theme included the two sensemaking factors that were linked to assumptions [9]. SM8 explored the assessment of performance assumptions and SM9 explored if the data changed assumptions. Alike the other questions related to sensemaking, the responses were varied. Whilst SM8 was a negatively worded question on the whole, most respondents felt as though the data helped them assess their performance (5% "strongly disagree" and 33% "disagree"). In relation to the final question (SM9) surrounding changing assumptions, the most common response was neutral (38% "neither agree or disagree"). This may indicate that practitioners had no assumptions before the reporting was shared and they could not be changed. Although this may indicate an openness to new initiatives, as practice analytics reporting becomes more integrated, perceptions on this may change. Practitioners may become more aware of their performance being reviewed in this way and have different assumptions about what they expect across different reporting periods.

5 CONCLUSION AND RESEARCHER REFLECTION

This case study explored the sensemaking of medical practitioners in the context of practice analytics. The results support the use of data to foster performance reflection as a form of CPD and present insights into the sensemaking process and considerations of how this can be supported. As referenced in the results, in order to ensure that the tools and reporting is designed around the end users sensemaking, the following human-centered research findings should be considered during implementation of practice analytics:

- When designing practice analytics tools, consider whether the audience views the data on individual or group level. Whilst both offer different benefits (e.g. tailored feedback and group discussion, respectively), we need to respect that practitioners have different preferences that will impact data engagement.
- Ensure that appropriate policy and implementation guidance is in place to protect the end-users' trust and acknowledge that reviewing data associated with individual clinical performance requires sensitivity and careful consideration. Developing a culture that accepts that data can provide meaningful insights can take time.
- Consider the use of "frames of reference" to help practitioners to scaffold the data and support them in curating meaning. Include the end-users within the design process to understand what comparisons are meaningful, and how to present them appropriately for the context (e.g. de-identification and level of granularity).
- Acknowledge that "cases of interest" offer opportunities to learn in ways that routine monitoring of clinical performance does not. Whilst summative data provides a holistic view of performance, if the performance is of a high quality, the

data may not generate any meaningful insights for learning or enhance practice as nothing may need to change. Create situations (e.g. highlight cases, create different visualisations, or drill-downs) that allow practitioners explore, interrogate, and question the data, and thus performance, from different angles.

- In order to be meaningful or lead to a change in practice, practitioners must attribute the data to their clinical performance; indicators must be developed in a way that reflects this.
- Be aware of different factors that impact practitioners' sensemaking within specific contexts (e.g. emotional affect, past experiences, and group dynamics) and attempt to respectfully account for these.
- Ensure that practitioners are included within design and implementation phases of practice analytics tools.
- Understand that sensemaking needs will change over time and that the development of reporting and practice analytic tools is both an iterative and a continuous process.

In general, practitioners felt that reviewing the data was an effective CPD activity and useful for learning and development. However, a challenge that practice analytics, regulators, and hospitals face, is how to “measure success”. Whilst practitioners (and hospitals) naturally (and appropriately) focus on patient outcomes and the direct impact an initiative has on the quality of care and practice, in the context of practice analytics and CPD, the impact that reflecting on data has on clinical performance could be considered a secondary outcome. As continually outlined, CPD aims to support practitioner development in a way that is relevant to their scope practice and grounded in evidence-based results [24]. Although the end goal is to ensure high quality practice, the process to achieve this is *through* learning and development. As acknowledged in the established field of learning analytics however, measuring “learning” faces its own challenges, even within traditional educational settings [11, 13], let alone in contexts not immediately associated with learning and development. Whilst it may be easier to measure the success of practice analytics through the impact it has on clinical outcomes, it is arguably not a valid measure of success. It does not capture the complex nuances of individual learning and development. For as long as we remain focused on leveraging the data to solely measure high level outcomes (and not drill-down to individual learning and development) we are not directly measuring learning, and the culture will not shift towards one conducive to learning and development that is required for effective sensemaking. Whilst hospitals may see “no high level change” in clinical outcomes after implementing practice analytics initiatives, they may be creating systemic changes to a culture that promotes learning, which a positive step-change for their practitioners. This positive, but nuanced, shift may not be captured in high level clinical outcome reporting.

In addition to the reflections above, given the varied responses provided throughout the results, it seems clear that a “one size fits all approach” is unlikely to be achieved in relation to sensemaking. First, consistent with other research [33], a blanket approach *can* not be achieved because the results demonstrate what is meaningful to one person will likely be different to next. Second, such approach

arguably *should not* be applied as it goes against the rich sensemaking perspectives that exist; regardless of research discipline or area, sensemaking is widely reported as an individual process that is shaped by our previous experiences and knowledge [20, 22, 29, 32]. As such, we must put the individual human at the centre of our exploration and not attempt to apply blanket nomothetic approaches to their individual needs. Future work should focus on such and leverage research approaches that account for these “human” individual differences (e.g. idiographic methods); a motivation of HCI research. Ultimately, good practitioners strive for the highest quality of care, if designed appropriately, they welcome the use of data to support them in achieving this. It is therefore important to ensure that practice analytics tools are designed *for humans, about humans*. Tools must support practitioner sensemaking so that they can translate these abstract data points into the humans and the livelihoods that they represent. That is what is meaningful to practitioners, and therefore meaningful for enhancing clinical practice.

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A.1 Methodological appendices

A.1.1 Quarterly report examples. Fig 1. shows an example of the summary data shared with practitioners each quarter (Format 1 referenced within the manuscript). This is aggregated on a practitioner level. Fig 2. shows an example of the planned procedure data shared with practitioners each quarter (Format 2 referenced within the manuscript). This is grouped by procedure type but aggregated for each practitioner. The 3rd format referenced in the manuscript is not pictured given the specificity (and therefore sensitivity) of information but is also presented within an Excel tab as a tabular list for each “case of interest”. Indicators include: same day procedures, average time to surgery, average theatre time, average length of stay, hospital acquired complications, return to operating theatre, transfusion rates, medical emergency team calls, intensive care unit days, use of robotic surgery, readmission rates, and mortality.

[illegible]

Figure 1: Example of the overview data shared with practitioners.

[illegible]

Figure 2: Example of the planned procedure data shared with practitioners.

A.1.2 Survey - Instrument. Below is a copy of the survey that was hosted in Qualtrics and shared with medical practitioners in conjunction with their quarterly report.

Section 1: Demographic questions

- (1) Are you a Physician or a Surgeon?
- (2) What is your speciality group?
- (3) How many years have you been clinically practising?
- (4) How confident are you extracting meaning from data?

Section 2: Sensemaking questions^{4 5}

- (5) I found that viewing the data made me reflect on my performance in a deeper way. (SM1, Reflection)
- (6) I found that viewing the data provided me with unexpected insights on my performance. (SM2, Insight)
- (7) I found that viewing the data made me really focus on different aspects of my performance. (SM3, Focus)
- (8) I found that viewing the data did not help me reconstruct and reenact my prior performance. (SM4, Argumentation)
- (9) I found that viewing the data helped me better identify areas to enhance the quality of my care. (SM5, Explanation)
- (10) I found that viewing the data provided me with ways to evaluate facts and evidence about my performance. (SM6, Assess Facts and Evidence)
- (11) I found that viewing the data did not help me to distinguish how I had performed. (SM7, Distinguish)
- (12) I found that viewing the data did not help me assess my assumptions about my performance. (SM8, Assess Assumptions)
- (13) I found that viewing the data changed some initial assumptions I had about my performance. (SM9, Change Assumptions)

Section 3: Feedback questions

- (14) How useful was this reporting for you?
- (15) I considered this data to be a valid reflection of my performance.
- (16) I found the data that compared me to my peers was interesting.
- (17) How comfortable would you be sharing your data with your colleagues?
- (18) Do you have any concerns about how this data may be used?
- (19) Do you think reflecting on this data should be recognised as part of your college Continuing Professional Development (CPD) credits?
- (20) Any and all feedback you have is useful to help inform future reporting. Please let us know below if you have any feedback on the quarterly data report that was shared with you. This can include any aspects of the reports (e.g. accuracy, presentation, access, indicators).

A.1.3 Interview protocols.

⁴The original scale explored *video debate* sensemaking, but the subject here was *performance data* sensemaking. To illustrate this adaptation, the first sensemaking question was updated from "I found that using the **website** made me reflect on **the debate** in a deeper way" to "I found that viewing the **data** made me reflect on **my performance** in a deeper way."

⁵All of the survey items (SM1-SM9) related to sensemaking were answered using a 5-point Likert scale (Strongly Disagree "1" - Strongly Agree "5").

Protocol 1: Currently clinically practising.

- (1) How do you define electronic health data?
Prompts: Any indicators that are taken from data? Any information stored on your patients or you? Any electronic data on patient outcomes?
- (2) Do you currently use any electronic health data?
- (3) How is this data currently presented to you and how often?
- (4) What experiences do you have with using electronic health data as a measure of clinical performance?
- (5) In the context of electronic health data, what types of data, tied to your personal clinical performance, do you currently receive feedback on?
- (6) How is this feedback currently provided to you?
Prompts: Visually or verbally? Paper or digitally? Access a portal or system?
- (7) How frequently do you receive feedback on your performance that has been derived from electronic health data?
- (8) Do you believe that this is an appropriate frequency and why?
- (9) Can you think of any information stored in electronic health records or other electronic systems that you would like to receive feedback on that you currently don't?
- (10) What do you believe are the benefits of using electronic health data to review your personal performance?
- (11) Do you receive any information that is a comparison of your personal performance and your peers?
Prompts: Anonymised group performance?, Colleague averages? Any other benchmarking measures?
- (12) What are your thoughts on receiving anonymised peer comparisons? I.e. you can see your performance and where this sits compared to your colleagues, but they are anonymised.
- (13) How is receiving this comparison beneficial or, why is it not beneficial?
- (14) After receiving feedback on your performance, how do you reflect on this? Is there anything you specifically do with this information?
Prompts: Reflect on the specific experience? Compare self to colleagues? Investigate the matrix / working out? Discuss with colleagues or other groups?
- (15) Do you receive this information in a meeting / discussion with your craft group or other group? If so, what do you discuss related to feedback, and what are your thoughts about discussing your performance with others?
- (16) How does reviewing these data impact your future decisions, clinical practice, or performance?
- (17) Can you please explain why you think that's the case?
- (18) Have you got any concerns about using electronic health data as a measure of performance, or as a feedback tool?
- (19) That concludes all the questions that I have prepared. Is there anything else that you would like to add related to the use of electronic health data as a measure of performance or as a feedback tool?

Protocol 2: Not currently clinically practising.

- (1) How do you define electronic health data?
Prompts: Any indicators that are taken from data? Any information stored on your patients or you? Any electronic data on patient outcomes?
- (2) What types of electronic health data do you currently use within your role?
- (3) What is your experience with using electronic health data to provide feedback on clinical performance?
- (4) Are you aware of any other types of electronic health data could be used to provide feedback to clinicians that is not currently provided?
- (5) Could you elaborate on how you access, store, and then distribute this information to clinicians?
- (6) What is your experiences with how this information is presented to clinicians?
Prompts: Graphs, tables, other visual prompts? Excel or other spreadsheet? Is the data interactive for the user?
- (7) What has been the feedback or reception from end users on providing this data as a measure of performance, or as a feedback tool?
- (8) What are your experiences with using electronic health data to create colleague comparisons or benchmarks, and are you aware of how this has been received by the clinicians?
- (9) In what ways do you believe providing this information impacts future clinical practice or patient outcomes?
- (10) What are your overall opinions on using electronic health data to review your personal clinical performance?
- (11) Are there any barriers to gaining this information or sharing it within your institution?
Prompts: Privacy or confidentiality issues? Any technology challenges? Issues with distributing information? Or any resistance from colleagues or organization?
- (12) That concludes all the questions that I have prepared, is there anything else that you would like to add related to the use of electronic health data, as a measure of performance or as a feedback tool?

A.1.4 *Survey demographics.* Table 2 presents the demographics of the practitioners who completed the survey.

A.2 Results appendices

A.2.1 *Response distributions.* Table 3 presents the response distributions for the sensemaking items within section 2 of the survey.

A.2.2 *Additional quotes to support themes.*

Data-driven practice reflection as a constituent of CPD.

- “this should be what the CPD is looking for” (survey)
- “the whole premise of having what we call a learning circle so that ... you contextualize those outcomes in terms of individual surgeons or the institution and then sometimes you modify practice or you modify data collection moving forward and continue the cycle again” (interview)
- “you have to collect what you think is most clinically meaningful, what’s most meaningful to outcomes” (interview)
- “so currently ... there’s very little summated data provided as a routine” (interview)

Performance reflection.

- “I think various clinicians find it variably confronting to see their own data compared to someone else. Particularly if they’re an outlier in terms of variance” (interview)
- “although I look at my own data and particularly reflect on outliers and complications, the data offers an overall picture of my performance and offers a time to reflect on it” (survey)

Generation of insights and information.

- “comparing LOS in particular and return to theatre is a good way to reflect my practice against others” (survey)
- “just to make sure I am not an outlier” (survey)
- “if I’m told my length of stays double everybody else, that would get my attention” (interview)
- “I found that data sharing is not universally open and honest - to openly share data, this needs to be done in an open, trustworthy environment.” (survey)
- “but for the audits, it’s pretty sort of standard you know everyone has got a few outliers and you know a few complications and we sort of work through them” (interview)
- “I would review it with the group and we would all look but I’d also use that to sort of bounce off any individual cases or outline cases sort of thing that would need to discuss. And that, and I think that would then inform any change in practice, or protocols that needed to be refined, or education that needed to happen” (interview)
- “I think it’s reasonable to send them specific data in comparison to other like doctors for instance ... but I think you get most value out of a group of discussion in that setting” (interview)

Performance reconstruction.

- “maybe smaller reports of a more focused/specific nature” (survey)
- “present the pooled summated data, but also allow the opportunity for doctors to present individual cases that might have been outliers to discuss ... common problems.” (interview)
- “so [specialist] community will look at it in a positive sense. So if you’re an outlier, you know, you’re not going to be seen as ... you’re not doing well. People will say well, what do I do to get back into normal?” (interview)
- “maybe smaller reports of a more focused/specific nature” (survey)

Data-driven evidence.

- “does not really assess the most important aspects of my work” (survey)
- “that was our first go and it was reasonably useful ... subsequently sent me more data ... I know that he was happier with the way that that was presented” (interview)
- “statistics used by companies are on the same sliding scale as lies” (survey)

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Table 2: Demographic breakdown of the medical practitioners that fully completed the survey.

Speciality Group	N	Median	Minimum	Maximum
Breast Surgery	2	20.5	3	38
Cardiothoracic	3	19.0	12	25
Colorectal Surgery	2	12.5	7	18
ENT Surgery	3	14.0	2	21
Gynaecology	1	31.0	31	31
Hepto/Biliary/Pancr	2	16.5	11	22
Neurosurgery	5	23.0	10	40
Obstetrics & Gynaecology	5	18.0	11	42
Ophthalmic Surgery	1	19.0	19	19
Orthopaedic Surgery	6	17.5	3	30
Plastic/Reconstructive Surgery	1	26.0	26	26
Upper GI Surgery	3	17.0	6	34
Urology	4	18.5	12	22
Vascular Surgery	1	16.0	16	16

Table 3: Response distributions and descriptive statistics for the nine sensemaking questions included within the survey.

Question	Strongly Disagree	Disagree	Neither Agree or Disagree	Agree	Strongly Agree	Median	Mean	Variance
Reflection (SM1)	10.26%	5.13%	28.21%	53.85%	2.56%	4	3.33	1.02
Insights (SM2)	12.82%	20.51%	43.59%	23.08%	0.00%	3	2.77	0.92
Focus (SM3)	10.26%	25.64%	35.90%	28.21%	0.00%	3	2.82	0.94
Argumentation (SM4)	2.56%	17.95%	35.90%	30.77%	12.82%	3	3.33	1.02
Explanation (SM5)	12.82%	12.82%	35.90%	38.46%	0.00%	3	3.00	1.05
Assess facts and evidence (SM6)	12.82%	7.69%	25.64%	51.28%	2.56%	4	3.23	1.18
Distinguish (SM7)	7.69%	28.21%	25.64%	23.08%	15.38%	3	3.10	1.46
Assess assumptions (SM8)	5.13%	33.33%	28.21%	20.51%	12.82%	3	3.03	1.29
Change assumptions (SM9)	12.82%	25.64%	38.46%	20.51%	2.56%	3	2.74	1.04